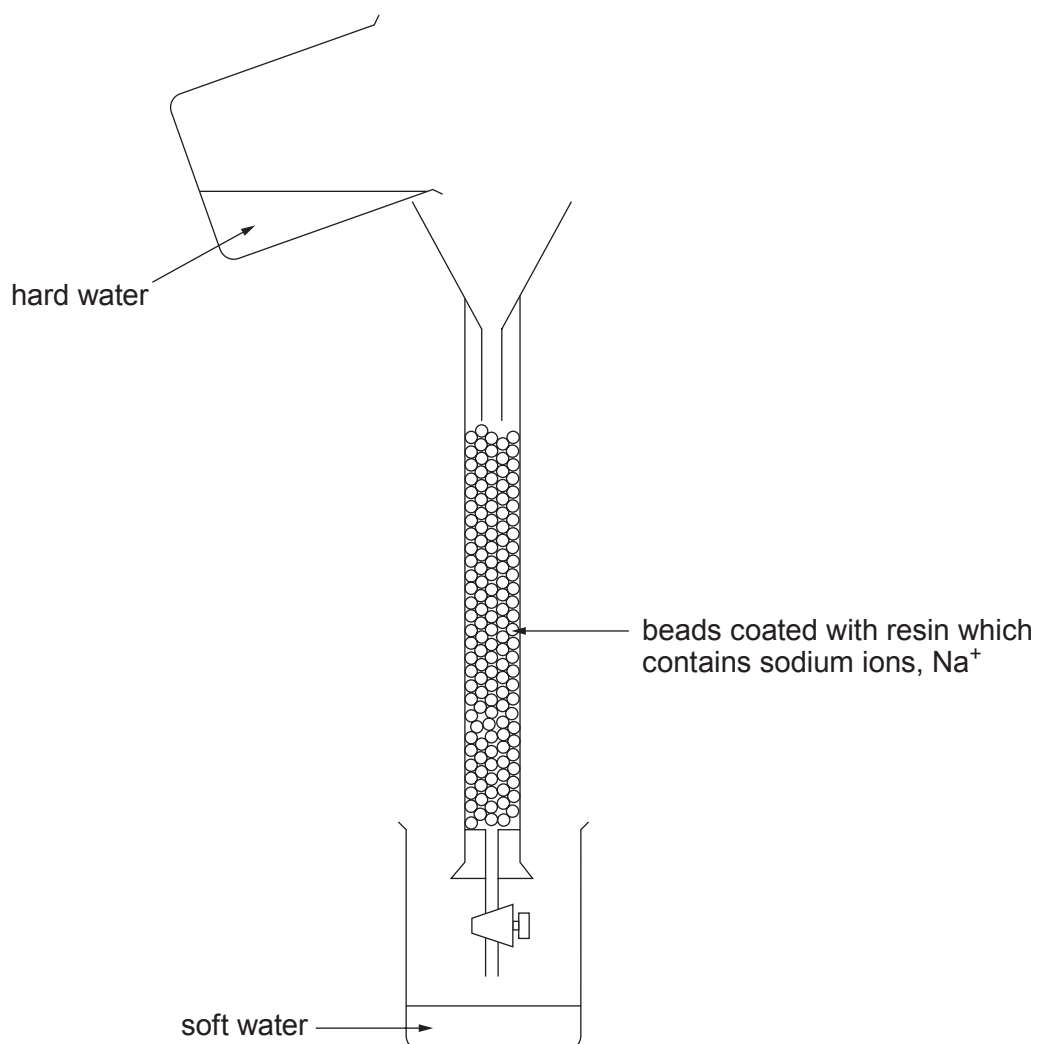


7. (a) The diagram shows an ion exchange column used for softening permanent hard water.



- (i) Explain how ion exchange works.

[2]

.....

.....

.....

- (ii) Explain why the ion exchange resin will stop working after continued use. Name a solution which can be passed through the resin to regenerate it.

[2]

.....

.....

.....

(b) Describe and explain a method to test whether a water sample is soft water, temporary hard water or permanent hard water. [4]

(You do **not** need to include reference to fair testing in your answer)

.....

.....

.....

.....

.....

.....

.....

.....

Examiner
only

8